

ASHLEY MNAKANDAFU

Computer Systems Engineer | Embedded Systems & IoT | Networking

Mutare, Zimbabwe (Open to relocation) | 0772 959 779 / 0781 973 270 | ashleymnakandafa@gmail.com

[LinkedIn: Ashley Mnakandafa](#) | Driver's License: Class 4 (2020)

CAREER OBJECTIVE

Results-driven Computer Systems Engineering graduate with hands-on experience in embedded systems, IoT, networking, and machine learning deployment. Proven ability to design and build real-world solutions – from an ADAS prototype on Raspberry Pi 5 to GSM-based industrial monitoring systems. Bring a creative, analytical mindset and a strong eagerness to learn, with the ability to approach problems from fresh perspectives. Seeking an IT role where I can apply my technical foundation and grow within a dynamic team.

EDUCATION

BSc Honours in Computer Systems Engineering – *Midlands State University* Graduated September 2025

- Class 2.1 – Strong performance across embedded systems, networking, and software engineering modules

CERTIFICATIONS

- Cisco Skills for All – Introduction to IoT (2024)
- Cisco Skills for All – Introduction to Data Science (2024)
- Cisco Skills for All – Introduction to Cybersecurity (2024)
- Cisco Skills for All – Cyber Threat Management (2024)
- Cisco Skills for All – Ethical Hacker (2024)
- HEXCO Certificate – IT Field (2017)

WORK EXPERIENCE

IT Intern (Industrial Attachment) – *High Court of Zimbabwe – Mutare* August 2023 – August 2024

IT Officer Duties

- Configured, installed and maintained network hardware including switches, routers and access points for a 100+ user environment.
- Provided daily IT support – resolving hardware faults, black-screen issues, software crashes and connectivity problems across court machines.
- Monitored network performance using Wireshark, SolarWinds and UniFi Controller; diagnosed and resolved connectivity issues.
- Supported 2–3 virtual court sessions every Tuesday and Thursday, ensuring stable AV and network connectivity.
- Performed network cabling, port labeling, IP address management and documentation updates.
- Assisted in configuring firewalls, VPNs and basic network security policies under supervision.

E-Filing Officer Duties (Voluntary)

- Guided lawyers and members of the public through creating IECMS accounts and filing court cases digitally.
- Provided frontline assistance at the e-filing desk, improving digital adoption and reducing filing errors.

Receptionist (Voluntary)

- Managed front-desk operations including visitor coordination, call routing and administrative support.

Sales Assistant & Informal IT Technician – *Retail Shop, Mutare* October 2025 – Present

- Diagnose and repair smart TVs, smartphones and personal computers for customers and walk-in clients
- Resolve home and small business network connectivity issues including router configuration and Wi-Fi troubleshooting

- Perform software installations, OS updates, antivirus setup and general device maintenance.
- Handle sales operations while doubling as the go-to technical resource for the business.

ACADEMIC & TECHNICAL PROJECTS

ADAS Prototype – Raspberry Pi 5 – *Final Year Project*

2024 – 2025

- Designed and developed a multi-functional Advanced Driver Assistance System with object detection, lane detection and real-time safety alerts
- Deployed a trained MLP model (TFLite) for ultrasonic-based object classification (pedestrian, car, motorbike, bus) and a 1D CNN for signal denoising
- Implemented speed-adaptive Forward Collision Warning (FCW) and Blind-Spot Detection (BSD) logic with a priority alert hierarchy: FCW > BSD > Lane > Engine
- Built engine thermal risk monitoring using DHT11 sensor data and a machine learning prediction model
- Integrated 16x2 LCD display, buzzer alerts, LED indicators and Bluetooth motor control for a full driver-feedback interface
- Technologies: Python, TensorFlow Lite, Raspberry Pi 5, Arduino Mega, ultrasonic sensors, IR sensors, DHT11

GSM-Based Industrial Fault Monitoring System – *Academic Project*

2024

- Built an IoT monitoring system that detects gas/smoke, abnormal temperature and unauthorized motion in industrial environments
- System sends automated SMS alerts to designated personnel in real time upon fault detection
- Integrated PIR motion sensor, MQ-series smoke sensor and DHT11 temperature sensor with an Arduino and SIM900L GSM module
- Displayed live sensor readings on a 16x2 LCD with threshold-based alerting logic
- Technologies: Arduino, C++, SIM900L GSM, DHT11, PIR sensor, LiquidCrystal I2C

TECHNICAL SKILLS

Programming Languages: C/C++ (primary), Python, Java (beginner)

Embedded & IoT: Raspberry Pi 3/4/5, Arduino Mega, GPIO, ultrasonic/IR/DHT11 sensors, SIM900L GSM, L298N motor driver

Machine Learning: TensorFlow Lite, MLP, 1D CNN, model training & edge deployment, Pandas, NumPy

Networking Tools: GNS3, Cisco Packet Tracer, SolarWinds, UniFi Controller, Wireshark, PuTTY

Databases: PostgreSQL, MySQL, Firebase, SQL Server

Developer Tools: Git, VSCode, JetBrains IDEs, Kaggle, Google Cloud Platform (basic)

LANGUAGES

- English – Fluent
- Shona – Fluent
- French – Basic

INTERESTS

- Embedded systems, robotics and automotive safety technology
- Data science and machine learning for real-world applications
- Chess (analytical thinking and strategic problem-solving)
- Keeping up with emerging technology trends

REFERENCES

Available on request, or contact below:

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